

ActInf Textbook Group ~ Cohort 7 ~ Session 21

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all right welcome everyone we're in cohort 7 the second discussion on chapter8 so there's a few ways we can go we can look at chapter eight in the context of the book um in terms of continuous time modeling uh following up on chapter 7 from discrete time modeling also uh after or just going to it I can share some updates on continuous time modeling from this morning in the RX and fur group where we um developed on the new example from the RX infer team with the lorenza tractor and an example with the neural network so we could talk about that and uh modify it in the context of continuous time modeling um first though let us just go to the chapter does anyone want to just share a section of chapter 8 or any question or quote or or image or anything like that I have a question okay Andrew then John uh yeah so they talk about um this attractor like this point that things tend to but they seem to talk about it on two different levels like one in reality and then one in the maybe the generative model I guess like and so if you could clarify that distinction um how those work and what's their relation yeah so when you specify the entire simulation you will have let's just say we're dealing with with some with temperature in the room we could have situation where just to to get into this Y is our observable X is going to be our latent State $\dot{x}$ is the rate the change landscape on the Latent state so this is like the thermometer measurements coming in on Y and the Latent State on X and the change in the temperature um you could have a room with a real attractor setting like there's there's actually a variable uh v such that when the true latent state is exactly V the the um rate of change is zero and when the room temperature is higher it drops back to the point of tractor and vice versa for being lower then you have the agents model of the external process that could be built to be structurally identical like they're both dealing with a point detractor in which point uh the agent model will tend to converge pretty simply to the correct parameters but you could also have a situation where like the real room is on a limit cycle so it's just oscillating or it's by stable but the agent's belief is that there's a point attractor so to answer it this is just one example of attractors there's Point attractors limit

attractors uh and so on and in practice you end up specifying the actual generative process that gives rise to the agent's observations and then you specify the agent's model of that process and those can be structurally identical or not so so the attractor is basically a construct of the a of the organism in the way that you're describing it like it's something it would wish for you know which may not exactly even exist there but may be useful nonetheless is that yeah I mean on on multiple levels there's there's the the modeler making the temperature model of the room and possibly just choosing to use an attractor style dynamical model just just as a map and then there's the agent within the simulation using her istics or constrained families to do variational inference in and and that could be simplified on up to any point and but the crucial thing is that is the attractor in the agent's mind that that there's something there they ch chasing like they're trying to tune into and at least of understanding and for me that's that's very uh relevant uh because I'm trying to do this like two mind uh dialogue where like you'll have a a continuous mind that's having some kind of affordance or ability and then the attractor would be like a concept and then that could feed into this like discreet mind let's say uh that you know has connections between Concepts let's say but so it seems like an attractor can serve as what I would call a concept and the process of approaching that would be like some kind of skill or ability or affordance well yeah I mean in figure 86 this is like a classic hierarchical architecture where the lower level is a continuous time continuous stat space model and the higher level is discreet time discreet State space model so it could be like you have some sort of continuous time Dynamic whether it's attractor or just some sort of oscillator and then it's like there's a discretization let's just say into quadrants so then even though there's this continuous two-dimensional State space at a lower level it gets discretized into just a four state discrete State space model that's basically digitization so this corresponds to a variety of settings it's really convenient for taking analog sensors and then discretizing the information it's been applied to folk Psych ology in terms of like embodied in interoceptive processes which are continuous in their nature leading to discretization and action selection okay good so that it has been done that's um okay cool um John and then yeah we can just go to any any part on eight or we can when we get to Len we'll look at kind of Len 2025 well well this is what I wanted to talk about actually because what I learned about the lorenza tractor or the luren system I should say I I I learned that it was a very simple system that isn't even very good for modeling the the weather uh and so I'm wondering how you could use this system in like all sorts of other uh other circumstances when it's not even that good at modeling the weather which is what it was made for yeah great great point so I'm not sure if it was actually meant to

be like the best possible weather forecast model meteorologist but the point was the simp it was just it was just meant to be a simple model that captured certain aspects of the system right and kind of the Breakthrough was that this deterministic dynamical model in continuous time exhibited chaotic Behavior so that showed that really simple models can't have complex chaotic I don't think that was a break through because um for example po K looked at systems like that around the turn of the 20th century uh the I think I think this though might have been the first use of the idea of an attractor but my question is if it's if it's not useful for what it was built for how can it be used to model even other things like um yeah yeah agreed Loren definitely is building on different attractor Concepts from po so let me just share a little bit from RX and fur um this were some some updates that that I pushed this morning but based upon um examples that they brought in just a couple hours ago so six hours ago um so again it it's not that the the Loren uh model has just a few parameters just here so if you wanted performance you might not necessarily use this exact model but conceptu I've have a weird problem with zoom I I I like I only see like my uh my window like in the main screen I can't see what you're sharing you someone help me with this maybe like right click on uh can you see the shared window at all right I I see say thing it says Daniel fredman screen but I click on it and it's very small and I can't read it you don't see it at all or no I I do see it but it's very small and I can't read it maybe do pin right click and pin it I'm try I'm right clicking but it's not doing anything um I'll try to figure this out yeah yeah yeah it's all good but just it is there so it has these several parameters and it's a it's a sort of proof of concept showing how with several parameters you can get this dynamic chaotic Behavior so what this RX Ander example did today so it's in RX Ander examples. JL in the advanced examples integrating neural networks with flux JL so they uploaded this six hours ago as a notebook which is really useful for like web-based viewing um and then I modified it into a script and made some modifications um so what what this script does we'll just look at it in the context of this notebook and then we can go over to cursor and and um play around with it um so first they Define the Loren's system this is going to be the underlying generative process so in in uh terms of Chapter eight so kind of go back and forth between chapter eight and this more recent work the underlying process is the X why are going to be the observations that are sampled from the Loren system a data set is created with noisy samples coming from this Loren system so here's this image this is kind of like the gerer boach book cover we have a three-dimensional continuous time dynamical system with Loren's Dynamic and then here are three of the projections onto each of the par wise axes so there's like we have some you know some insect flying around in a

three-dimensional room and these are the projections with a shadow on three of the walls um so we are using this because the noisiness and the underlying chaotic nature of Lorenz system make it hard to find a convergent parametric estimate of the transition Dynamics so it's emblematic of complex adaptive systems like weather or biological systems that might dwell in a regime and then have some kind of switching into another regime again showing how that comes out from just a couple of um of equations which are deterministic but the sampling from them is not any chaotic system is going to be like that right so so why Lorenz yeah because it's classic and there are other ones this is one with very very few parameters so like there's the double pendulum there's a variety of chaotic systems to study like empirically and analytically and this more of a metaphor no we're actually studying this model what do you mean well um if you have a chaotic system and you're representing it with the Lorenz system like that that wouldn't be good unless it actually is modeled by the Lorenz system right well you can have a toy model right like you can study a toy model just for lots of reasons yeah there's the pure pedagogical element which is like if you don't have this Lego block in your grammar you won't have a key piece of understanding how it works in this case and then if there's a different model that fits the data better when you're in like a data fitting situation like chapter nine right you might use a different model or so is the reason you use Lorenz because it data fits is that the answer is that the answer is no because it's like has a long history it's simple and it gives us an intuition on these chaotic switching Dynamics with a lot of um like rich behaviors from a simple underlying just a couple equations which we can define in a few lines of code okay but so then there's like a world of active inference that works with chaotic models besides Lorenz and Lorenz is just in the textbook is that is that the thing yeah you like if you put a different model here the rest of the notebook you could just this is just an example so if you have a different model or you want to change how you can put a different model here just wait to understand what's what's happening in this notebook so a simple model just keeps things short and easier to like Daniel said if it's classic then people understand there it's easier to communicate yeah there's so and it is one of the simplest I mean it's just these three dimensions and three updates okay so here's noisy samples coming from Lorenz tractor now we want to infer the lat and state given the observed observations so flipping to chapter eight that we are getting observables and we want to infer neural states or latent states like in SPM statistical parametric mapping so kind of the precedent to FP and active inference you'd have like neuroimaging data as your observables and you have underlying neural activity as latent states okay however the hidden dynamics of the Lorenz system exhibit nonlinearities and cannot

be solved in the closed form one manner of solving the problem is introducing a neural network to approximate the transition Matrix of the Loren system so note that the notation is different in this example than it is from the textbook so in this example A is going to be the transition operator and B is going to be like the ambiguity operator that Maps between the latent States and the observations whereas in the textbook it's like the other way around B is the transition states in the latent space and A is the observation Mountain okay flux. JL is used as a neural network package in Julia here's the RX andur probabilistic model we could return more to this or or people can learn more about RX and fur or get involved in the project but this is a very simple State space model with just five variables um first the probabilistic model is used to extract parameters from the untrained neural network and basically it does terribly like the untrained neural network doesn't predict the data at all um so it was just randomly initialized and so the probabilistic state space model also just this green line it it has nothing to do with the data then the neural network is trained using a free energy objective which is how neural networks are trained um here's just neural network training code then we get here which is showing that we can use the probabilistic model the state space model like the explicitly defined RX and fur model to recapitulate the parameters of the neural network and so I took that several steps further um it's super super exciting because this is showing a a really critical design pattern which is about fusing the complimentary strengths and functionalities of the empirical data fitting highly parameterized nature of like machine learning and neural networks with explicitly defined and constrained probabilistic models like in RX and fur so basically they have complimentary strengths neural networks are Universal function approximators they can interpolate any given function you know given enough training and size and all of that um however they have high computational resources they have challenges with interpretability and so on they don't have genuine uncertainty estimators all of these different kinds of limitations here's showing a a way in which the parameters of the lorens can be inferred with a neural network and trained on the data just like anything else in machine learning or just sort of computer science today and then we can infer transition parameters within a very very tightly defined probabilistic model which is small enough to be defined within this app model Block in RX and fur so this is kind of it's It's you know things have happened since 2022 this is one of them from just a couple hours ago um and it's just really exciting because it shows like you can do pure endtoend probabilistic programming on base graphs to directly fit parameters of distributions from data and and that may be adequate that might might be direct and awesome but this is a super interesting pattern that can be added

into a lot of extent machine learning pipelines which is or language model pipelines where you have some kind of reinforcement learning or neural network system and then you use explicit probabilistic modeling on top of that to gain interpretability genuine uncertainty estimators and other kinds of benefits like lower compute resources for these probabilistic models John uh so uh are you or is anyone familiar with step wolfram's concept of computational irreducibility yeah so um you you said that the the AI model can match any function how does does that work with computational irreducibility or uh are we rejecting computational irreducibility what's how does match I said interpolator or approximator so yeah like you could have let's just whatever the actual ontological computational irreducibility is of like the weather you can train a linear model to approximate it it might be a terrible approximator your startup might fail you might pick the wrong weather you know Etc ET Etc but you can approximate with a model so those Blobs of irreducibility in nature can be approximated using statistical models so like to take this into sort of like the halting problem like you can't know whether a script is a computer script is going to like what its output is going to be for some kinds of scripts but you could have a statistical estimator that says I I think there's an 80% chance this script is going to Halt without running it or you know you human behavior could have these elements of computation or irreducibility but that doesn't mean you couldn't make a statistical model as a behavioral scientist so it's kind of like map territory like irreducibility is about the territory but you can always make statistical approximators but it does it create a limit on how good the approximation can be right like if the approximator becomes perfect then functionally there was computational reducibility because you could reduce the underlying process to something simpler but to the extent that the approximation is not perfect then that computation has not been reduced now whether it's in principle irreducible or just the modeler hasn't tried hard enough is a second question but like here's here's an example of the advantages that you get from building this probabilistic model so here is like on a given Dimension so just for the x value or something of the loren attractor we get a a moving posterior distribution with a mean and a variance but the neural network won't give us a mean and a variance it's just giving us its best prediction of the location at the next step whereas this gives us like a moving bayesian incredible interval so the these these kinds of methods are are complimentary um and it I mean just again within the last hours this has greatly expanded our toolkit because it gives us a access to building probabilistic models on top of trained machine learning models rather than needing to do endtoend training only through the probabilistic model so could you maybe expand on that Point like where you said with the mean and the

variance that uh I guess you're getting this uh data coming in from a neural network and then you're able to you said I think statistically approximate that from the other end like from top down or yeah yeah so like let's just say that that we have a language model and we're going to ask it um how many people live in x Place MH and it is going to do this forward pass through billions of parameters and then it's going to say like 11 million 10.5 million 13 million 17 million like it just it's outputting these numbers which represents in that run with that temperature the likeliest outcome that's how it works with sampling the likeliest outcome given some some noise that's added back in we could take that sequence of observations and then fit a model that just fits a mean and a variance from the language model's outputs so we kind of infer like the true value for its belief about the population of that region from sampling its outputs so it still could be wrong relative to the real reality like let's just say all of its training data suggested that there's this amount but then more people are there so this the this isn't to say that it's veridical in some future moments but inferring the latent State and having an uncertainty estimator on the lat and state is always going to be more accurate than any given single measurement it's like the measurement of you know if we have the mean and the variance of the height of the children in the classroom that's always a better um number to work off of than just one measurements and and I think that's a nice example of what this whole setup is that the statistics kind of makes this presumption that there's something meaningful that we call the mean and the and the variance but the reality might be that this this neural network is just all over the place and that's just there's no really I mean it that means nothing but from in from a statistical point of view in terms of corralling or shepherding or like just trying to you know get get your H our hams around this it's very useful and it is practically use ful even though it from a meaning point of view it might not actually mean anything but it suggests what a meaningful meaning would be like that if it was if this is a meaning you're supposing hey like is this what you're talking about like is this what you mean the machine would say I don't know what you're talking about but that's how I'm thinking about it yeah so MB you mentioned balance let's just see if cursor will uh will bless us today okay what's kind of a question that we could ask about balance and see if we could be dealing with almost this two minds Paradigm of like a neural network that's fitting noisy proprioceptive data from a continuous dat space and then like a conscious explicit mean and variance estimation from there but like MB was an example of a setting like that or what would be a question that that relates to the vestibular Coke CER yeah um C could you help me understand that a bit better as I I'm I'm yeah trying to follow uh and and yeah you can just call

me Mark um okay okay yeah mb's cool for Markov blanket too oh yeah sorry I didn't see that my uh my screen name is that Markov blanket um Yeah so basically like using um cursor as sort of this this um augmented coding setup just it it may not work within you know the the 30 minutes we have here but just just to see possibly um that's what these examples are great for is like starting with this structure and then modifying tweaking the example like the neural network could be flipped out for a more sophisticated neural network or here in this situation let's see if we could adapt it to study some question of Interest with like a vestibular clear balance question yeah so are we sorry um specifically so we're trying to put data into this model is that right we're trying to build a model using this this program is that correct yeah yeah modify this the the architecture of this model where we get noisy data from a continuous dat space you know chapter 8 and then we're going to infer the we're going to use a neural network to train on the parameters from this noisy State space that could be likened to our subconscious um you know embodied predictive networks and then infer parameters from a probabilistic model put on top of that neural network okay so if we're looking at some kind of uh data like um and tell me if I'm if I'm understanding this correctly like postural sway is that would that be one of the data that we're looking okay okay the noisy data are coming from proprioceptive continuous State spaces related to postural sway and and that and that could be like modeled as Loren attractor with with um chaotic transitions between qualitatively different postures like right the threedimensional stat space could be um the three angles of your finger so then there's like a Zone within that state bace that the finger the finger's biomechanic ICS are related so it's like and this again to kind of get to this like what's what's the distinction with active inference and these other neural network or other kinds of models what would be the approach to making biomechanically realistic finger movements well one approach would be we get hundreds of hours of movement of fingers and try to do endtoend learning based upon recapitulating realistic looking videos of hands for example um but obviously even with billions of hours of video making images with realistic hands is still challenging whereas this complimentary approach would be you actually have this explicit parametrically defined model of the biomechanics of the hand and then you understand well here's the zones of State space within that defined model that the hand can be in and then will generate images of the hand from these biomechanically realistic postures versus just trying to recapitulate the the appearance from real video and so so okay and then what what what kinds of question would we want to address with that kind of model so are you I mean I could say something like you know one's uh conscious perception of threat you know some something

like that would would that be is it fair to so this is all new to me right but but taking someone's uh you know conscious perceptions um and looking at the effect on on the other end of the you know entropic dimension of their postural sway characteristics you know some something like that that that would be I I don't know if that's the the right kind of question to to ask for for this type of exercise but you know that would be an example of something that would be of interest I'd like to add a related example uh learned about uh when I went with my friend to his son's aido practice yesterday and there was a um he told me about ban Jean so there's this martial arts it's all about these different hand movements I there like eight different ones so like they just cycle through them and so you can um so it's very much like these attractors like you can think okay my hand's going to be grasping or my hands going to be blocking let's say or my hands going to be striking and so in each case I kind of picture I think they even kind of suggest that that there like if your hands grasping you have a particular attractor that you're projecting and so as you move your hand you this this attractor is guiding the grasping so that in that context you would you would be imagining this grasping and and you would be centered on that and it would all work out but then in this Kung in this B ban School based on the iching eight trigrams then you might go to let's say striking you see and then you might go to let's say blocking so the context keeps changing for what you what is an attractor and you just C cycling through all these attractors it's it was like bot periodicity I talk to Daniel sometimes about but I thought that was kind of curious that wonder if like that could be connected here yeah like um this was uh it it relates to chapter chapter eight like in this page on uh this is we we did a book stream with Thomas par a couple years ago and uh this part right here relates to basically recurrent attractors and continuous State space model so initially the active inference models were more like generalized filtering then people wanted these sequential Dynamics so that's like dealt with in chapter eight like cursive so then it was like there was models where it was like the limit attractor was like a cursive letter s so it's just kind of like drawing a cursive letter S and the continuous Dynamics were basically like the arms joint mechanics on the table um but planning on digital computers is much easier with discrete State space models and so that is what led to these hybrid architectures where there are mixed continuous and discrete State spaces because planning is is more easily considered in the discretized setting like a chess planner um so it's like with the eight hand postures um it's like that there's eight discrete options at this higher level and then action selection you know context switching attention which is modeled as internal or covert action meaning it's not observable from the outside attentional regime shifts within this discrete State space um is you

a descending motor contact switch such that the body engages in a different limit attractor so it's like raise your leg now put your leg down now raise your leg now put your leg down so it's like there's a biomechanical continuous attractor in that case it would be basically like that would be the example from here move move your point okay point at the top corner of the room not point at the bottom corner of the room not Point At You Know point over there then there's this transient Dynamic new set point is established and then um error error um reduction with propri receptive feedback no touch your right wrist and then well once you do that that the proceptive error is zero when you have when you're when you're doing that the top down attentional regime I should be expecting Pro receptive feedback reflecting that I'm touching my wrist and the ascending propri receptive data have no error term and that's a fixed point or the fixed Point could be oscillatory like wave your arm like this so that's like an oscillatory ATT tractor one thing I've I've come to understand is that as we so so use the example of moving your arm to a point in order to initiate actually to plan and conduct that movement there has to be sensory I'm assuming that's propri receptive attenuation is is that right and so how do you then use the appropriate receptive input to understand the movement in the model when I think what we do is and you can absolutely correct me is we turn the Precision down as we attenuate the sensory in to in order to to overcome the the evidence right that that I'm not moving to use that example does that make sense yeah sensory continuation comes up in five chapter 5 as well but yeah exactly like modulating attention so what are the two ways to to reduce free energy there's change your mind and change the world you know updating beliefs that's changing the Mind through learning an inference or updating action to to to modify the world okay and what is going to control whether the agent engages in passively updating its parameters now parameter updating is its own kind of action but not gross action you know or changing through through action and that relates to the attention that's paid to the ascending predictions basically so that is is empirically observed in postural settings as well as like Cades which is like kind of like micro postural I mean it's it's ocular postural um that basically in order to enable a move if you kept High Precision like for example all my perceptive is that I'm seated in a chair so how is it that I can get out of a chair when it's like no I and and in reinforcement or reward learning actions are selected based upon their expected reward whereas in active inference we're doing direct policy inference on the likelihood of action so it's not that standing up would be more rewarding for me it's I expect myself to be standing up so it's the Fulfillment of likelihood rather than the maximization of reward but how could it be when it's an obvious counterfactual how could it be likely for me

to be standing up how could I be the kind of guy who's standing up when my vertical appropriate reception is that I'm not and the answer is transient sensory attenuation makes it suppressing the proprioceptive input enabling there to be more ambiguity of which position I'm actually in which creates a kind of like postural liminality which then a top-down belief no actually I am standing up or I'm on the path to standing up can sort of get the ball rolling over a critical threshold that then action fulfillment can take all the way that's been modeled in like situations with like hyper or hypo dopaminergic cases where like somebody may have challenges initiating movement but once movement is initiated it continues or they can initiate but not complete a movement so it's just that transient suppression I think that was that's the key is that it to initiate and then you're on to some level processing that proprioceptive feedback to to help with the movement itself right when the movements in process there can't be this continuous suppression it has to be just I picture it just kind of as a as a Flicker and then you're gradient a gradient of suppression like yeah well said you become more and more precise demanding and more certain and more precise yeah or or it's like driving on a windy road and it's like on a straight part you want to accelerate but then you relax off in order to update your vector so it's like when your gaze is fixed you want to be having high attention to the visual input but then during intentional or sub intentional eye movement visual attenuation occurs which is why we don't get motion sickness you know don't get nauseous outside of like vertigo type settings with eye movements because when we look away like we get this attentional blink we're actually not accepting or paying attention to the visual input even though photons hit the retina the same rate during the movement of the eyes so that's like kind of like it's like depressing the clutch on attention shifting into a different spot and then we re-engage the Precision and now we're like getting information from over here I say there's good evidence for those who for folks who go on to um have you know something like persistent dizziness is that there is not uh good uh sensory attenuation with movement in fact there's there's this excessive Precision that seems to be a theme among you know a variety of uh conditions yeah that like that is sort of like that is what Precision Psychiatry and all these kinds of approaches get towards which is like sometimes it's that there's a um it can be an uncertainty estimator that's too precise or not precise enough that leads to different behaviors so it's not a question of just like moving a variable up or down if I was to refine that question that that you kindly asked me to try to formulate it say something like um were the effects of excessive tension on postural sway you know which would be the same thing that would probably occur during a you know a perception of threat um but what are the effects on the actual sway parameters so

this just becomes this ongoing Loop and that would be you know something like to to use the you know the tibular phraseology like Mal debarkment which is just like an excessive um duration of something like SE legs yeah I don't know if that's something we could put into a model like this yeah like the it's that's kind of that that is the exploratory modeling and inquiry is like okay so we have this model that's getting trained on Pro receptive data so you know in progress but then we could say something like well but let's just say somebody is acting based upon their parametric estimators of posture what would happen if their if the Precision was low so let's just red let's have it let's have the the best fitting model and then let's simulate movement from someone whose Precision is three times higher or three times lower and then that you know yeah Andrew did you have a yeah I wanted to jump in with an idea that may actually be related to this but that I've been having in this situation where let's say you have this task that you're given a triangle and you're going to put in uh and you're kind of blind but you're going to put in a little circle and you're try to expand the circle until its maximum size and you can never be perfect at that but you you try what you can so you're given some triangle so in the beginning when you do a task like that you're going to be bumping into sides or whatever like you don't know how to guess what would be the right Maxim you know the right amount to increase the radius Etc but at a certain point you'll get the answer or something rather satisfactory and then I imagine there's like a replay in let's say in your mind to say look okay this is what it took for me to do this task but now that I know what the answer is where the center is I'm interested to learn from that by saying okay now that I know where I end up I would replay it in my mind to do that as efficiently as possible like I'd make the right guesses all along the way and so if you do the Replay for the cases you've done before um you get much more efficient algorithms that you're developing so let's say you're getting you're given a whole bunch of different triangles and so they're different conditions and whatever but I think this notion of and I think it goes like to the statistical um interfacing that you were doing Daniel saying okay it's not a particular answer but like you're going to get a statistical mean and a variance for like what's optimal Behavior you know what's optimal presumption like where is this center of the triangle probably most likely going to be if I'm if I'm thrown into some kind of triangle yeah yeah like live this notion of Replay that you you would want to replay and get your optimal strategy and then you'd want to go into a new problem with that optimal strategy and keep improving it yeah so like live stream 51 series with isura at all this is a really critical paper that shows that there's a monotonic relationship between the loss function used in neural network fitting and the variational gradient descent on the semantics of

the base graph across the blanket and they use this postdictive learning strategy where basically at each time point you minimize so the factor graph with fictive causality of factors MH so decision making to minimize the future risk so we want to minimize future risk for the unoccurred event and the way that's going to happen is by remembering our past sequence of actions and observations and then reassessing what would have been the least regretful updated risk assessments given all that we've seen till now or given within the sliding window of what we've seen till now so then it's fictive causality is propagating backwards in time in the sense of the the agent is saying if I would have done that given what I know about this now this is how I think it would have occurred and using those re-entries to reduce upcoming risk and it can be moment like very rapidly after the final solution so like with posture I can just imagine like let's say you feel wobbly and you kind of like regain your balance but you just say wait wait wait like I had it all wrong like this is what I should have been having in my mind so because so and if you don't have that you're just going to be wobbling all the time and you're going to fall down but if you're wobbling a little bit and you're able to reparameterize well what is the my proo SPC situation then you'll be you'll feel stable that was just a momentary thing that was somebody said said something about about what I was just saying you could you could also do that on on a false inference right like um like an overattribution of something like let's say you you have a really and we see this in the clinic you have really kind of fluky circumstances around a fall you know that you couldn't possibly you know recreate sort of a oneoff experience that somebody has and then they go through the process of it sounds like mitigating the likelihood that that will happen again and they do that to a level that is excessive right so then their subsequent movements become very constrained like shuffling their feet or something yeah as for example or they have a very like tight I guess you'd say um you know Excursion of of their of their postural sway you know so so those kinds of things and and they're same time there's a greater conscious and this is what I'm interested in is like you know the the the conscious over tuning of [Music] precision yeah like the I mean that is is definitely where nested models would come into play like an injury occurs and then there's increased attention to to the pain or there's modification and that sets up a new attention with descending and appropriate exception ascending and those can be those can result in a new attractor W with reduced Mobility or whatever it may be and that can generalize to to other postures and and even probably other circumstances yeah like again that's sort of the toolkit and that's where the textbook gets us it's just like hey with with single layer models this is the there's already these richness of cognitive patterns to explore

and that's what chapter 7 even and chapter eight are about so you know chapter six is the recipe these are the questions to consider like going through these is is qualitative initially and then chapter seven goes into the discrete time models chapter eight goes into continuous time models chapter 9 goes into Data fitting but it's like that that kind of is one structure of the process and then it's maybe you want the single best performing model for your deployment or or for your paper or or there's also a lot of value in building these portfolios of models and looking how different features like modify uh you know explain different components of the variant especially if it's going to be something involving people where there's like intersections of of complex psychological phenomena so it's it it it may not it may not be just a single layer model but this is some of the toolkit and knowing what what a single layer model can do helps give give some of the um motifs kind of like the rudiments for building more sophisticated models and I just it's an exciting day because the applicability of these probabilistic models to do parametric inference from neural network models which are just everywhere has increased markedly in the last like eight hours which is pretty fun to jump in like to think of these two minds one continuous and one discrete and I think of like the continuous mind you know you want to extract the critical point so it could be like the Peaks or the troughs or it could be like you know going up your tensing your muscle you know and then going down you're relaxing it and so you have these different modes that then can become discrete and I guess what I learned today from you was that in the other direction it's like projecting statistical uh presumptions like oh there's got to be a mean let's say there's got to be a variance that's attracting even if they don't actually you know exist in any real uh physical sense but they exist in a sense of trying to get a grip on what's what you you know getting a handle yeah I mean and and and this this like to the early questions like the structure of the generative model assumed by the agent does not have to be the same as the environments the environment could be like a bathtub probability distribution like it's either like dwelling low or dwelling high but then the agent could be trying to infer a gaussian so then they could having all these pathologies where first it's super tight low and then it's super tight high and then it's in the middle and it's super blurry and it's swapping between them or it's staying in one so it's like it's like they're fitting Ultra well when it's on the low with their High Precision low estimate but then they're completely confused when it flips to the high state so it's like those are those are the kinds of Diagnostics that you want to look at to understand whether you're dealing with the wrong family of variational distributions and it's like you could be walking uphill for a ways and then walking downhill for the ways like your posture is complet like there's no average

that make it's a it's a non-concept to say there's some kind of average there because depends on where you're walking and you don't know where you're going to be walking but but the idea that you have some kind of momentary balancing you know parameters that you're kind of pre guessing like that's they don't have to be real but they could be meaningful I think yeah and and it comes back again to the uh Precision modulation and attention mhm which is just over the the papers and the years is is always returned to as sort of a a Hallmark of intelligence you know you want to be precise when it's good to be precise and to be open when it's good to be open but when to do that is the question and that's that's like a third mind type of thing balancing those two minds I think exactly agreed with that um just in our last minutes let's see how that um what this what it will do for its second output but anyone have any last comments or questions I have a question but it's not about chapter eight is it is it possible to change your generative model like are there there models that change models or yeah yeah that's called structure learning is when the topology of the model the structure of the model is itself modeled so yes then so any other uh I'm glad John Mark Daniel that to hear from everybody was very helpful today thank you cool thank you for uh allowing me to participate and and learn this is this is great cool yeah I'm also just really curious to see where where this last training step will get us in the outputs I mean just and I hope like again it's not all about code but starting with examples and modifying them and kind of staying on a ridge of functionality is super powerful right now like you can also this RX INF for you're active in this or um it's an open source project we have a pro we meet weekly okay you are in this m i mean it's lazy Dynamics and uh reactive bays are the developers but like there's a lot of participants in The Institute who also contribute and so and we're partnered with them but it just using cursor or similar methods again so it isn't all about code it's not all about reducing it to code but I this shows a little bit and there's there's other places to see it as well you can get you with one prompt you can get pretty far but modifying what's already functional is ridiculously powerful so that's one thing that's oh sorry sorry goad I was just saying it's becoming a bit demystified as far as that that was the one question I had coming in is okay how do you take something that's conceptual and then take real numbers and and run it and it sounds like it's it's it's kind of like well you you just start with something and then you say okay what if I increase the Precision here or I change the model there right that's kind of what we're getting at is you just start to apply numbers in a in a relative sense is that right if if the and if even better if there are real data but I don't know how you would give um you know Consciousness you know how you would give that numbers you know like like attention you just were playing with that zero

to one as far as the Precision of within the model is that what you're doing yeah I think like obviously Consciousness especially if it's defined as unobservable is is a perennial measurement issue in question um doing attentional modeling and more of these like um just cognitive variables that aren't necessarily related to self-awareness or or a Consciousness those are more tractable self-report is a classic um so yeah Consciousness related modeling is often tried and very um fraught but but this is I I don't mind it being FR so this distinction between the continuous and the discrete and then realizing like you were saying uh Mark that sometimes you shouldn't overthink it right like you should be uh doing you know your continuous movements sometimes you sh it's just like your lucky mind when everything is going well just keep walking but you have the Unlucky mind like hey there's a hole like don't fall in right so so do you use your lucky mind or your unlucky mind um uh that is what Consciousness is all about like to make sure they have the same information and then deciding yeah but here I shouldn't overthink it but oh no here I should be careful so I think and see to to the things we're learning like the relation between those two minds one is that like oh let go of the break in terms of certainty and pretend you don't know and then tighten tighten tighten Titan so that shows you that process shows you the direction of those two minds that you're going from The Continuous one that needs to be moving to the discrete one that needs to kind of like focus on some attractor so those types of things are indicators of um the Consciousness in play I would say let's see if it will output the stick figure as requested but but you know these functions are only a couple uh so this could be like this is pointing and like you're having your hand waver and point at something and then like for example blood sugar is related to the Precision of the ability to point if you're if if somebody's having a a blood sugar crisis they're going to have different Precision of their movements that's something that's more empirically observable able and then you could differentially parameterize more overt elements let's see if it it if it did it absolutely hilarious these are stick figures yeah true posture propri receptive perceived posture so that gets super interesting right like there's the real body and then there's like these ghosts of like what we predict and what is and then you just say okay now modify it so that it's taking in this data instead of the simulated Loren data so say which which what are the three the true the true posture is the red one the proprioceptive signals are these noisier gray ones okay and then the perceived posture is the blue one that's that's exactly what people describe is you know at some point okay I realize my postural sway as normal and that that almost takes some convincing and then but it feels like I'm you know I'm doing this it feel and you've probably had that experience you know um coming off a boat or you even sometimes

transiently coming off an elevator you you perceive that excess and for some people there seems to be a difficulty attenuating that and and you know returning and and to use your uh example like you're walking up a hill and then you're walking flat but like when you made that transition you're still in a sense like running the program as if you're walking up the hill and now you have this this conflict yeah that's the that's fascinating looking at that that that's really exciting a really great way to uh to Tom yeah it's open source take it and and run slw walk with it not too fast or stand and sway yeah epic though it's epic okay thank you all see you later bye thanks so much great bye bye